Yale Invitational - Repaired practice edition - Answer key

Division C | Disease Detectives | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

4. (2 points) Authored replacement for source item Question 4: A diagnostic test gives 3 true positives and 7 false negatives among 10 infected patients. Calculate the false-negative rate as a percentage of infected patients.

answer: 70 % (2 points).

Numerical target 70; absolute tolerance 0.14. Accepted units: , %.

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

reasoning: An onset curve has three bins with counts 6, 8, and 2.
A report-date curve would incorrectly look like 16 simultaneous reports if interpreted as onset.
Use onset timing to generate source hypotheses and assess completeness before claiming the source. (3 points).

conclusion: The time variable changes the meaning of the curve. (3 points).